

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT **Paper -2(Adv-2024-P2-Model)** **Date: 07-09-2025**

Time: 02.00Pm to 05.00Pm **GTA-01** **Max. Marks: 180**

07-09-2025_Sr.Super60_NUCLEUS-BT_Jee-Adv(2024-P2)_GTA-01_Syllabus,

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | |

For More Join: @IITJEE_Advanced

JEE-ADVANCED-2024-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023


For More Join: @IITJEE_Advanced

MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. Let $f(x)=\begin{cases} x-1; & x=\text{Even} \\ 2x; & x=\text{Odd} \end{cases}$, where $x\in N$. If for some $a\in N, f\left(f\left(f(a)\right)\right)=1$ and $m\&n$ respectively are the number of points in $(-a,a)$ at which $g(x)=\left|x^2-1\right|$ is not continuous and not differentiable, then $m+n$ is

A) 0 **B)** 2 **C)** 4 **D)** 6

2. Consider two circles in a plane with radii 5 and 3 respectively. Distance between centres of these two circles is 10. If R_1 is radius of circle passing through 4 points of contact of direct common tangents of these two circles and R_2 is radius of circle passing through 4 points of transverse common tangents then $\frac{R_1}{R_2}=$

A) $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$ **B) 2** **C)** $\frac{5}{3}$ **D)** $\frac{25}{9}$

3. For $x,y,z\in[0,2\pi]$. The number of ordered triplets (x,y,z) satisfying

$16\left(\sin^4 x\right)+2\left(\sin^4 y\right)+4\left(\sin^4 z\right)-16(\sin x)(\sin y)(\sin z)+2=0.$

A) 20 **B)** 18 **C)** 16 **D)** 14

4. If every root of a polynomial equation $h(x)=0,$ of degree '5' with leading coefficients '1' is real and distinct, then the number of real roots of the equation $h''(x).h(x)-\left(h'(x)\right)^2=0$ is

A) 0 **B)** 2 **C)** 3 **D)** 1

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 YEARS 1984-2025
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA Sri Chaitanya 300
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY Sri Chaitanya 341 360
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI Sri Chaitanya 720
For More Join: @IITJEE_Advanced

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Let z be complex number such that $|z|=1$ then possible values of $|1-z|+|1+z^2|$ among the following is/are

A) $\frac{1}{\sqrt{2}}$ **B)** $\sqrt{2}$ **C)** π **D)** 4

6. $\lim\limits_{x\to 0}\left(\frac{1}{x^{2024}}\int\limits_{x^{2025}}^{x^{2024}}\left(\frac{(1+t)^{\frac{1}{2023}}-1}{t}\right)dt\right)=\frac{1}{P}$. Then which of the following is/are correct?

A) Remainder when P^P divided by 35 is 7

B) $\ell n\, x = P\, x$ has exactly two real solutions

C) $(2020)^x+(2021)^x+(2022)^x=(P)^x$ has no real root.

D) Unit digit of P^P is 7

7. Consider

$$f(x)=\begin{vmatrix}5 & 7 & 3\\ 2 & 9g^2(x) & 2x^2\\ x & 3g(x)-x^2 & 1\end{vmatrix}-\begin{vmatrix}2 & 4 & 5\\ 2x^2 & 2 & 9g^2(x)\\ 1 & x & 3g(x)-x^2\end{vmatrix}+\begin{vmatrix}2 & x & 0\\ 9g^2(x) & 3g(x)-x^2 & -2\\ 2x^2 & 1 & -1\end{vmatrix}$$

(where $g(x)$ is polynomial function) such that $m=\int\limits_0^1 f(x)\,dx$ is minimum. Then

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 YEARS 1984-2025
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

A) $m = 1/5$

B) $m = 2/5$

C) Number of point of inflection of $\phi(x) = \int_0^x \left[\left(\int_0^t g(y)\,dy - \left(e^t - e^{-t} \right) \right) \right] dt$ is 0.

D) $g(x) \le \cos x \forall x \in [-1,1]$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and theon-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:+4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

8. Let f be a differentiable function satisfying $\int_0^{f(x)} f^{-1}(t)\,dt - \int_0^x (\cos t - f(t))\,dt = 0$ and $f\left(\frac{\pi}{2}\right) = \frac{2}{\pi}$ then $\left[\int_0^{\pi/2} f(x)\,dx \right]$ is _____. Here $[.]$ denotes G.I.F.

9. If A and B are two matrices of order 3×3 such that $2A + 3BB^T = 1$ and $B^{-1} = A^T$, where A^T represents transpose of matrix A also $T_r(A)$ represents trace of matrix A and $\det(A)$ represents determinant of matrix A, then $T_r\left(A^{-1} - 3B^3 - BA\right) + \det\left(A^{-1} - 3B^3 + BA\right) =$

10. Let $f(x) = 2x^3 - 3x^2 - x + \frac{3}{2}$ and $\int_{\frac{1}{8}}^{\frac{7}{8}} f(f(x))\,dx = \frac{m}{n}$ (where m, n are relative prime) then value of $(m+n)$ is equal to _____

11. Let the sequence is 4,7,1,8,9,7,6,…… for $n > 2$. The nth term of the sequence is units digit of the sum of the two previous terms. Let S_n denote the sum of the first n terms of this sequence. The smallest value of n for which $S_n > 10004$ is k, then $(k - 1990)$ is __

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 Years 1984-2025
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA sri chaitanya 11th-12th Class 300
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY sri chaitanya 11th-12th Class 341 360
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI sri chaitanya 11th-12th Class 720
For More Join: @IITJEE_Advanced

12. If the number of non-empty subsets of $\{1,2,3,4,5,.....,2000\}$ the sum of whose elements is divisible by 5 is 'n'. If $5(n+1)=2^{\alpha}\left(2^{\beta}+1\right)$ then value of $\alpha+\beta$ is

13. If the volume of the parallelopiped having three coterminous edges as $\vec{a},\vec{b}$ and $\vec{c}$ is '5', then volume of parallelopiped having three coterminous edges as

$\overline{\alpha}=(\overline{a}.\overline{a})\overline{a}+(\overline{a}.\overline{b})\overline{b}+(\overline{a}.\overline{c})\overline{c}$

$\overline{\beta}=(\overline{a}.\overline{b})\overline{a}+(\overline{b}.\overline{b})\overline{b}+(\overline{b}.\overline{c})\overline{c}$ and $\overline{\gamma}=(\overline{a}.\overline{c})\overline{a}+(\overline{b}.\overline{c})\overline{b}+(\overline{c}.\overline{c})\overline{c}$ is

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreenvirtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +3**If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Let $X=\{1,2,3\}$. A number is selected from the set X with replacement and this process is repeated 1100 times. For each $i, i\in\{1,2,3\}$, let $f(i)$ represent the number of times the number 'i' is selected. Also, let S denote the total sum of 1100 numbers selected. If $S^3=162f(1).f(2).f(3)$, answer the following questions.

14. The harmonic mean of $f(1),f(2),f(3)$ is equal to _____.

15. There are $f(1)$ red balls, $f(2)$ blue balls, $f(3)$ white balls and 'x' green balls (balls of the same color are identical). The balls are arranged in a row such that no two balls of the same color are consecutive. Let x_1 and x_2 be the minimum and maximum values of x respectively for which the above arrangement is possible. The value of x_1+x_2 is equal to ____.

PARAGRAPH-II

A person repeatedly tosses a fair dice. He gets two points for a throw of a perfect square and one point otherwise. Let P_n denote the probability that he reaches a score of exactly n.

16. The value of $\dfrac{P_{10}-P_9}{P_8-P_9}=$

17. The value of $\lim\limits_{n\to\infty}P_n=$

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 sri chaitanya 300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 sri chaitanya 341 360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 sri chaitanya 720
For More Join: @IITJEE_Advanced

PHYSICS **Max Marks: 60**

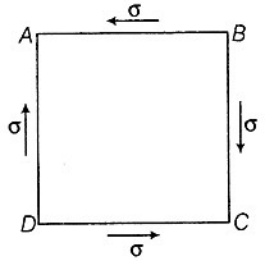
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

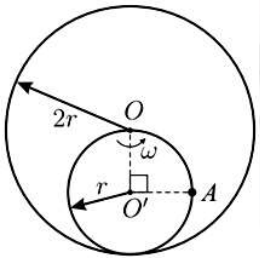
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

18. The figure illustrates a square block labelled $ABCD$, with a thickness of one unit perpendicular to the plane depicted. Each of the four faces experiences a pure shearing stress of magnitude σ, as indicated in the figure. The plate remains in equilibrium under the specified stresses. The highest normal stress experienced by any plane is:
(where the plane in question is perpendicular to the ABCD plane)



A) σ **B)** $\sqrt{2}\sigma$ **C)** 2σ **D)** $\frac{\sigma}{\sqrt{2}}$

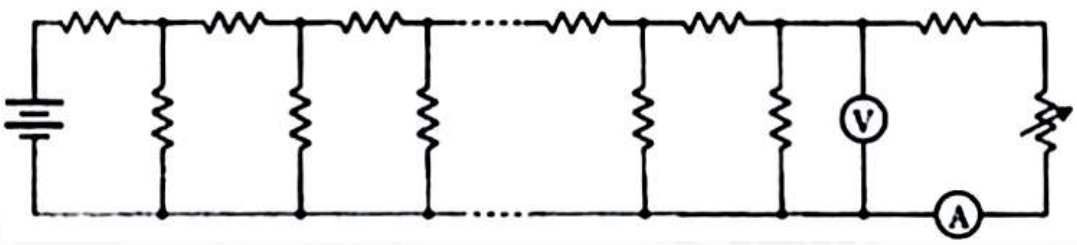
19. A small hoop, having a radius of r, rolls along the inner frictionless surface of a larger, fixed stationary hoop with a radius of $2r$. Both hoops have horizontal axes. At this particular instant, the smaller hoop is at its lowest position, and the segment OO' that connects the centers of the hoops rotates anticlockwise with an angular velocity ω. The angular velocity of smaller hoop is also ω anticlockwise. The acceleration of point A on the smaller hoop at this instant (Point A is situated on the horizontal diameter of the smaller hoop) is:



A) $\omega^2 r$ **B)** $\sqrt{2}\omega^2 r$ **C)** $2\sqrt{2}\omega^2 r$ **D)** $2\omega^2 r$

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 years
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

20. The network shown consists of an ideal battery, an ideal voltmeter V, an ideal ammeter A, several resistors of different values and a rheostat. If resistance of the rheostat is increased by a finite amount, how will the readings of the voltmeter and ammeter change?



A) Both will decrease

B) Voltmeter reading will decrease and ammeter reading will increase.

C) Voltmeter reading will increase and ammeter reading will decrease.

D) Voltmeter reading will not change but ammeter reading will decrease.

21. A long cylindrical wire of radius ' a ' carries a current i distributed uniformly over its cross-section. If the magnetic field at distance r and R from the axis have equal magnitude, then :-

A) $a=\dfrac{R+r}{2}$ **B)** $a=\sqrt{Rr}$ **C)** $a=\mathrm{Rr}/\left(R+r\right)$ **D)** $a=R^2/r$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

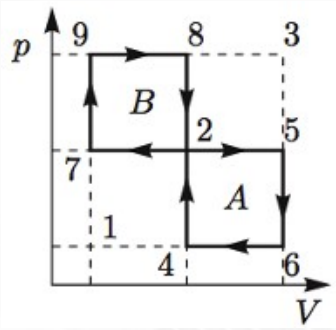
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 YEARS 1984-2025
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

22. An ideal monatomic gas performs a cyclic process A consisting of two isochores and two isobars. The same gas then performs a similar process B (Fig.). Let the efficiency of process A and cyclic process B be to be given and equal to η_A and η_B. In both processes $\Delta p_{21} = \Delta p_{32} = \Delta p$ and $\Delta V_{21} = \Delta V_{32} = \Delta V$, their numerical values are unknown.



A) The change in internal energy in cyclic process A is equal to the change in internal energy in cyclic process B.

B) Heat absorbed by the gas system in cyclic process B is larger than that in cyclic process A.

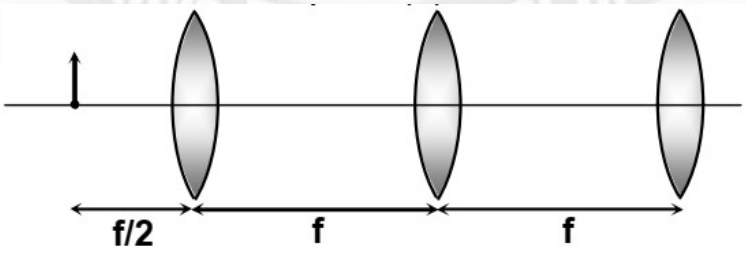
C)The relation of efficiency of cyclic process A and cyclic process B are given as

$$\eta_B = \frac{\eta_A}{1+\eta_A}$$

D)The relation of efficiency of cyclic process A and cyclic process B are given as

$$\eta_A = \frac{\eta_B}{1+\eta_B}$$

23. Three identical biconvex lenses of focal length f are aligned with two neighbouring lenses separated by a distance f as shown in the figure. A small object is located at a distance $f/2$ in front of the leftmost lens. Choose the correct option(s).



Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341 360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
For More Join: @IITJEE_Advanced

A) The final image formed is real and inverted.

B) The final image formed is virtual and erect.

C) The final image is of same size as that of object.

D) The final image formed is virtual and inverted.

24. Two metal balls of densities 7gm/cc (Neodymium) and 10.5 gm/c (Silver) are balanced on beam balance with different arms lengths. When the balls are completely lowered into the water, the balance is not disturbed. Then pick up the correct statement(s).

A) There has to be a cavity in silver ball if there is no cavity in neodymium ball.

B) There has to be a cavity in neodymium ball if there is no cavity in silver ball.

C) The volume of cavity in neodymium ball is equal to one third of the volume of neodymium ball if silver ball has no cavity.

D) The volume of cavity in silver ball is equal to one third of the volume of silver ball if neodymium ball has no cavity.

SECTION-III
(NON-NEGATIVE INTEGER)

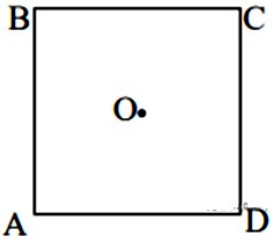
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and theon-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

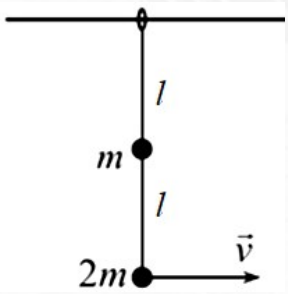
25. Consider a nuclear reaction $A + B \rightarrow C$. Nucleus A moving with kinetic energy of 5 MeV collides with a nucleus B moving with kinetic energy of 3 MeV and forms a nucleus C in excited state. Find the kinetic energy of nucleus C (in MeV) just after its formation if it is formed in a state with excitation energy 10.65 MeV . Take masses of nuclei of A,B and C as 25.0, 10.0 and 34.995 amu. $\left(\text{Take } \ 1\text{amu} \approx 930\text{MeV} / \text{c}^2\right)$

26. A uniform square plate $ABCD$ with side l moves so that it remains in one plane all the time. It is known that at some moment in time the magnitudes of accelerations of vertices A, B and C are respectively equal to $a_A = 5m/s^2, a_B = 7m/s^2, a_C = 5m/s^2$. Then find the magnitude of acceleration of the vertex D in m/s^2.

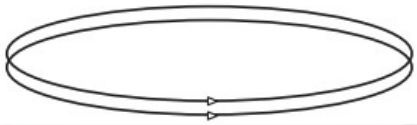
Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 Years 1984-2024
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA 300
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY 341 360
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI 720
For More Join: @IITJEE_Advanced



27. A very light ring can slide without friction along a fixed horizontal rod. Two small bodies of masses m and $2m$ are attached to the ring by two weightless, inextensible threads of length l (see figure). The system is initially at rest. The minimum horizontal velocity that must be imparted to the lower body so that during the subsequent movement the bodies can end up at the same height is equal to $v=\sqrt{Pgl}$. The acceleration due to gravity is g. Fill the value of P.



28. Two identical superconducting rings are coaxially placed very close (infinitesimally small separation) to each other. Each of the coils has of inductance L and carries a current I_0 in the same direction. If the minimum work done in making their separation infinitely large is $\frac{P}{2}LI_0^2$, the Fill the value of P?



29. A positively charged particle travels through uniform electric and magnetic fields that are perpendicular to each other. At one point in time, its velocity is represented as $\vec{v}_0$, with $\vec{v}_0$ being perpendicular to both $\vec{E}$ and $\vec{B}$. Assuming that $E=v_0B$, let the velocity of particle after sometime be $\vec{v}=-P\vec{v}_0$ (when its velocity vector makes an angle of 180° with $\vec{v}_0$). P is a dimensionless number, fill the value of P.

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 YEARS 1984-2024
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

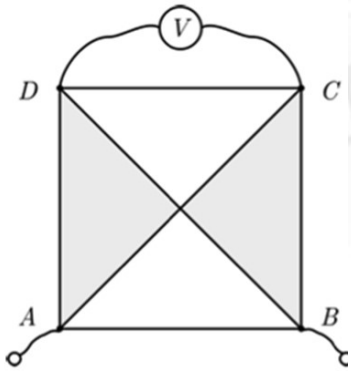
30. A smooth inextensible string of length ℓ is tied at supports A and B which are at the same horizontal level and separated by a distance $\frac{\ell}{2}$. A small bead is released from A to move on the string without friction. The speed of the bead immediately after the string becomes taut is $\sqrt{\frac{g\ell\tan 37^0}{n}}$. Find the value of n.

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +3** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

A square plate is made up of two types of conductors: grey and white (see figure). The resistivity of a white conductor is half that of a grey conductor. The resistance of the plate between the vertices A is equal B to r_1. If a current I is passed through these vertices , then the ideal voltmeter connected to the vertices C and D, shows the value of the voltage U_1. Now the resistivity of white conductors is made half the previous value, and of grey conductors is made 1/8 times its previous value. Now the resistance of the plate between the vertices A and B is equal to r_2, and when the same current I was passed through these vertices with the same voltmeter, the value began to show the value U_2.



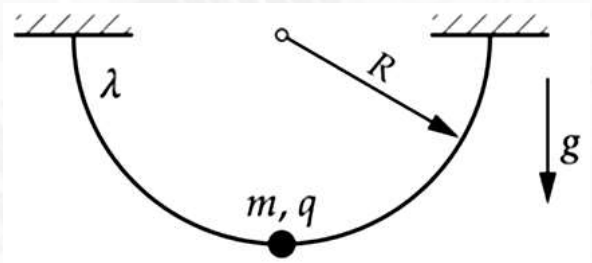
Sri Chaitanya Educational Institutions
Infinity Learn
40
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

31. The resistance of the plate between the vertices B and C before resistivity changes were made is Pr_2 Fill in the value of P.

32. The resistance of the plate between the vertices A and C before resistivity changes were made is $R_{AC}=\frac{1}{2}(r_1+Pr_2)+\frac{U_1+QU_2}{2I}$ .Fill the value of P+Q.

PARAGRAPH-II

There is a uniformly charged smooth half ring, whose upper ends are rigidly fixed such that it remains stationary in vertical plane. The charge per unit length on half ring of radius R is λ. There is a charged small bead at the lower position of ring having charge q and mass m. For all questions below take the shift of bead from lowest position to be very small as compared to radius of the ring. Take $k=\frac{1}{4\pi\varepsilon_o}$.



33. The value of charge q for which the equilibrium be stable is given by $q\le\frac{\sqrt{P}\,mgR}{3k\lambda}$. Then the value of P is____.

34. The time period of small oscillation of bead is given by $T=2\pi\sqrt{\frac{\sqrt{\mathrm{P}}mR^2}{\sqrt{\mathrm{Q}}mgR-kq\lambda}}$. Find the value of P+Q_____.

Sri Chaitanya Educational Institutions
Infinity Learn
40
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341 360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
For More Join: @IITJEE_Advanced

CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

35. Calculate the resonance energy of gaseous benzene from the following data.

$BE(C-H)=416\,kJ\,mol^{-1}, BE(C-C)=331\,kJ\,mol^{-1}, BE(C=C)=591\,kJ\,mol^{-1}$,

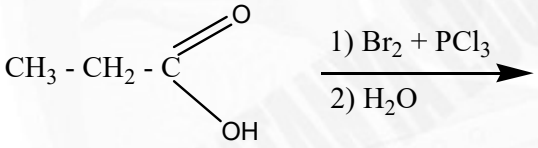
(BE =bond energy)

$\Delta_{sub}(C,\text{graphite})=718\,kJ\,mol^{-1}, \Delta_{diss}(H_2,g)=436\,kJ\,mol^{-1}$,

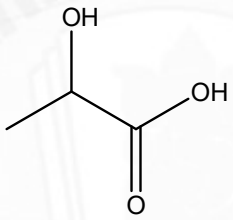
$\Delta H_f(\text{benzene},g)=83\,kJ\,mol^{-1}$

A)82 KJ/mol **B)**271KJ/mol **C)**352 KJ/mol **D)**200KJ/mol

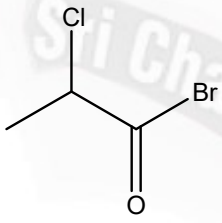
36.



Correct statement about the above reaction is



A) Major product is



B) is one of the intermediates involved.

C) In place of PCl_3 if red P_4 used then also major product of the reaction not changed.

D) In place of Br_2 if Cl_2 used then also major product of the reaction not changed.

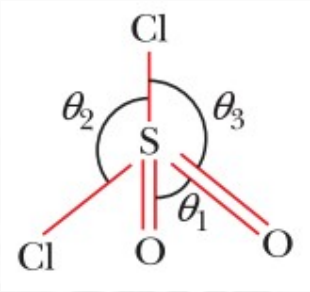
Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 YEARS 1984-2024
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA sri chaitanya 300
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY sri chaitanya 341 360
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI sri chaitanya 720
For More Join: @IITJEE_Advanced

37. In the chlorination of CH_4 in the presence of sun light which of the following is one of the chain propagation steps?

A) $CH_4 + {}^{\bullet}Cl \rightarrow CH_3Cl + H^{\bullet}$ **B)** $Cl_2 + H^{\bullet} \rightarrow HCl + Cl^{\bullet}$

C) $CH_4 + Cl^{\bullet} \rightarrow {}^{\bullet}CH_3 + HCl$ **D)** $Cl_2 \rightarrow 2Cl^{\bullet}$

38. Which of the following statements is correct for



A) It contains $p_\pi - p_\pi$ and $p_\pi - d_\pi$

B) It has regular tetrahedral geometry.

C) $\theta_1 > \theta_3$

D)Plane which contains maximum number of atom is 4.

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. From the below Latimer diagrams, which statement(s) is/are correct?

In Acid Solution $\left(\left[H^+\right] = 1M\right)$

$$ClO_4^- \xrightarrow{E^0=+1.19V} ClO_3^- \xrightarrow{E^0=+1.21V} ClO_2^- \xrightarrow{E^0=+1.65V} ClO^- \xrightarrow{E^0=+1.61V} \frac{1}{2}Cl_{2(g)} \xrightarrow{E^0=+1.36V} Cl^-$$

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341 360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
For More Join: @IITJEE_Advanced

In Basic Solution $\left(\left[OH^-\right]=1M\right)$

$$E^0=+XV$$

$$ClO_4^- \xrightarrow{E^0=+0.36V} ClO_3^- \xrightarrow{E^0=+0.33V} ClO_2 \xrightarrow{E^0=+0.66V} ClO^- \xrightarrow{E^0=+0.40V} \frac{1}{2}Cl_{2(g)} \xrightarrow{E^0=+1.36V} Cl^-$$

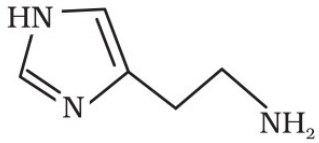
A) 'X' value is 0.542V

B) In acidic medium Cl_2 can undergo disproportion

C) In acidic medium ClO_3^-, ClO_2^- can undergo disproportion

D) In basic medium ClO_2 can undergo disproportion

40. Correct statement about histamine is/are



Histamine

A) It is formed when Amino acid Histidine undergo decarboxylation.

B) In Dumas method, weight percentage of nitrogen in histamine cannot be estimated

C) Interaction of histamine with the receptor proteins in the stomach walls can be prevented by the tegamet.

D) Agonist like brompheniramine is used as antihistamine.

41. Which of the following complexes is/are square planar

A) $\left(AgF_4\right)^-$ **B)** $\left(AuCl_4\right)^{1-}$

C) $\left[NiCl_2\left(P\left(CH_3\right)_3\right)_2\right]$ **D)** $\left[NiCl_2\left(P\left(C_6H_5\right)_3\right)_2\right]$

SECTION-III
(NON-NEGATIVE INTEGER)

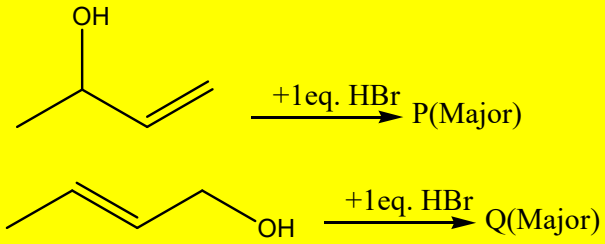
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and theon-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

42. Number of possible $\alpha - D -$ aldohexopyranose(s) is/ are X. $\frac{X}{4}$ value is

43. Allyl alcohols on reaction with HBr gives allyl bromide

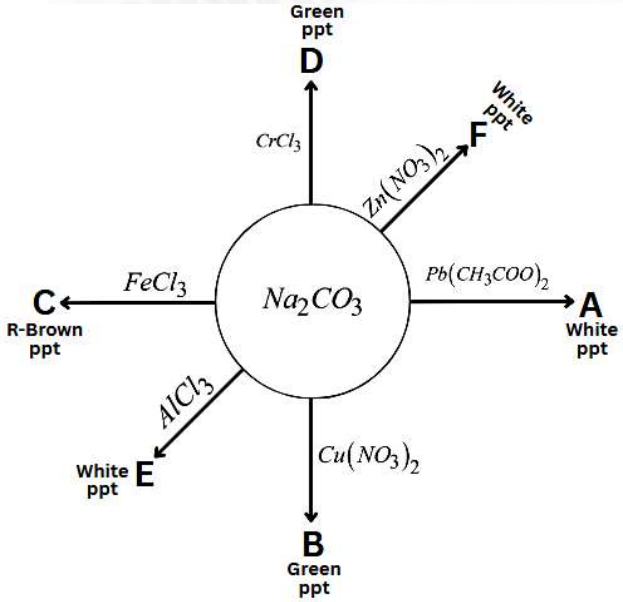


In P and Q, sum of the locant of Br as per IUPAC name is

44. Mixture of liquid A and liquid B is an ideal solution. Vapour pressure of pure liquid A and pure liquid B are 100 torr and 900 torr respectively. One mole of liquid A and one mole of liquid B are mixed. Find out the vapour pressure (in torr) when one mole of mixture has been vaporized.

45. In a sample hydrogen atoms and He^+ ions are present at fourth excited state in both. Maximum number of possible spectral lines in the emission line spectrum for the sample are

46.



The number of hydroxy or hydrated oxide precipitates are equals to

Sri Chaitanya Educational Institutions
Infinity Learn
40
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

47. The maximum number of atoms (including 'F' atoms) that may lie in the same plane in $(CH_3)_2C=SF_2(CH_3)_2$ is _____.

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +3** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

HA is weak acid, BOH is weak base. Ionization constant (K_a) of HA is 3×10^{-7} and ionization constant (K_b) of BOH is 3×10^{-7} at 298K.

48. Percentage of hydrolysis of 0.1M BA salt at 298K is

49. $HA+BOH \rightleftharpoons BA+H_2O; K_c=P\times10^{-Q}$ (In correct scientific notation). $(P+Q)$ value is

PARAGRAPH-II

Silicon forms a very large number of compounds containing SiO_4^{4-} anion as the basic unit. The structure of this basic unit is a tetrahedron in which oxygen atoms are arranged in a tetrahedral around the silicon atom.

(atomic weights of Si=28, Mg=24, O=16, H=1, Al=27, Be=9)

50. The percentage weight of silicon in the sheet silicate $Mg_3(OH)_2[Si_xO_{10}]$ is ____.

51. The weight percentage of unshared oxygen (univalent charge) in the total weight of oxygen in beryl $[Be_3Al_2Si_xO_{18}]$ is ____.

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
40 YEARS 1984-2024
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA sri chaitanya 6th-12th Class 300
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY sri chaitanya 6th-12th Class 341 360 MARKS
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI sri chaitanya 6th-12th Class 720
For More Join: @IITJEE_Advanced

Sri Chaitanya Educational Institutions | Infinity Learn

Sri Chaitanya Techno School
The right mentor for IIT (JEE), NEET, Olympiad & all Other Competitive exams

40 Years of Excellence 1986-2025

JEE MAIN 2025

31 STUDENTS BELOW 100 AIR



VANGALA AJAY REDDY
APP. NO. 250310255592
CLASSROOM STUDENT FROM GRADE IX - XII



DEVDUTTA MAJHI
APP. NO. 250310016185
DLP/AITS STUDENT



TOSHNIWAL SHIVEN
APP. NO. 250310391420
DLP/AITS STUDENT



SAKSHAM JINDAL
APP. NO. 250310236696
DLP/AITS STUDENT

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 10 ALL INDIA CATEGORY RANKS COUNT	BELOW 100 ALL INDIA CATEGORY RANKS COUNT	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT
31	95	10	98	579

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT

Sri Chaitanya Tops JEE ADVANCED

ALL INDIA OPEN CATEGORY RANKS



RUTVIK SAI
H.T.No. 256055278 (OBC-NCL)



MAJID MUJAHID HUSAIN
H.T.No. 251134112*



UJJWAL KESARI
H.T.No. 252016104*



AKSHAT KUMAR CHAURASIA
H.T.No. 254065055*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

*DLP/AITS

For More Join: @IITJEE_Advanced